Spatiotemporal Forecasting of Cellular Growth Dynamics using ConvLSTM-GAN Framework for Sequential Time-Lapse Microscopy

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***Abstract*—Predictive modeling of cell growth dynamics becomes a necessary component for understanding cellular behavior, which leads to drug discovery and regenerative medicine. It provides a novel approach for forecasting cell growth patterns using temporal sequence learning with Convolutional Long Term Short Memory(ConvLSTM) and Generative Adversarial Networks(GANs). The LiveCell Dataset was used,containing high resolution time lapse microscopy images of Neuroblastoma cells exibhiting cell differentiation for 3 days, each with 3 different locations,to train models that can predict the differentiated growth state of the cell in the next time interval using historic sequenced images. ConvLSTM processes spatiotemporal features to generate accurate single frame predictions, while the GAN framework provides realism to the image and biological plausibility of generated growth patterns.The model generates probabilistic growth maps that identify regions with high cell growth potential, with predictions naturally exhibiting soft boundaries that reflect biological uncertainty . Experimental results demonstrate that our model achieve a precision of 0.763 and SSIM of 0.0601 and average IoU of 48% with actual cell growth. Interpretability analysis points out that the different time windows from the very beginning do not have the same predictive power, as Frame 1 is making a cell growth forecasting contribution that is significantly more than the rest of the frames. The present study lays the groundwork for AI prediction of cellular behavior that can be a source of pharmaceutical research, cancer research, regenerative biology, and developmental biology.**

***Index Terms*—ConvLSTM, GAN, Cell Growth Forecasting, Spatiotemporal Modeling, Microscopy Image Analysis, Temporal Sequence Prediction, SSIM, Occlusion Sensitivity**

I. Introduction

Prediction of cellular development over time remains the main challenge of computational biology, thus cancer research, tissue engineering, and drug discovery are the most impacted areas. Traditional methods—like manually observing through a microscope and using statistical models—are often inadequate in recognizing the complex spatial and temporal aspects of cell division and interaction.

By means of Convolutional long short-term memory (ConvLSTM) networks and generative adversarial networks (GANs) temporal imaging data can be analyzed to find a possible solution to the problem. By combining spatial feature extraction with temporal sequence modeling, the methods used thus can become very accurate in predicting cell movements, shape changes, and interactions. The biological data-driven models that result from this not only enhance the predictive capabilities in different biological scenarios but also expose the cellular mechanisms that govern the behavior.

The LiveCell dataset is an excellent source for this task because it provides time-lapse microscopy images depicting cell growth, division, and communication.But predicting what will happen to cells in the future .States derived from historical observations pose distinct challenges: Cells show complicated changes in their shape, growth patterns frequently exhibit non-linearity, and biological processes inherently possess stochastic characteristics. These problems are tackled in this work by using two different but complementary methods: (1) a ConvLSTM-based regression model that directly predicts future cell states from temporal sequences, and (2) a GAN framework that learns the underlying distribution of cell growth patterns to create biologically plausible future states. Our contributions

- A novel spatiotemporal architecture for cell growth forecasting
- Occurance of a region in the image showing cell regrowth
- Interpretability analysis revealing temporal importance patterns
- Quantitative evaluation of forecasting accuracy on the LiveCell dataset
- Insights into model behavior through occlusion analysis and feature importance mapping

II. Related Work

The LiveCell dataset has been widely used to perform cell segmentation, enabling quantification of cell count, morphology, and spatial parameters using CNN-based architectures such as U-Net [1], [15]. These methods effectively delineate cellular regions but primarily focus on analyzing present cellular states rather than predicting future dynamics. This work bridges that gap by repurposing the LiveCell dataset for temporal forecasting, extending beyond static segmentation.

Traditional computer vision techniques [4], [16] and, more recently, deep learning architectures have been employed to predict cell behavior. U-Net variants [3], [18] remain dominant in cell segmentation and region tracking, while recurrent models have shown potential in temporal modeling. However, the majority of these studies target segmentation and tracking in the current visual field, with limited emphasis on forecasting future cellular states [17].

The work that is most similar to ours in that it employs CNN classifiers to predict the induced pluripotent stem cell growth status from time-lapse images is the approach that is most closely related to ours [11]. Their approach leads to the prediction of iPS cell growth with an 85% accuracy two days in advance when cells are divided into high, medium, and low growth groups. Apart from showing the possibility of growth prediction, their method suffers from a number of drawbacks that our work is designed to overcome:

Categorical vs. Continuous: They determine discrete growth categories, whereas this ConvLstm-GAN model produces continuous pixel-wise growth probability mapsGlobal vs. Local: They give the overall growth status, whereas this project pinpoints the exact areas of regenerative potentialClassification vs. Regression They implement classification frameworks, whereas spatiotemporal regression with ConvLSTM are used.This work moves from categorical classification to spatially accurate forecasting of growth dynamics thus being able to provide not only quantitative predictions, but also biological interpretability.

ConvLSTM networks [2] have been a great choice to figure out issues which are space and time dependent, e.g., meteorological forecasting or the next frame prediction in a video. The reason behind their success is that they combine convolutional layers with LSTM units, hence, they can detect spatial patterns and at the same time be informed of the temporal changes, thus, they are the best instrument for timelapse cell imaging. Unfortunately, only a few papers have acknowledged the potential of ConvLSTMs in tackling biological image sequences [5], [9], and the predominant theme of those research articles has been on classification rather than growth prediction.

One of the most significant advances for Generative Adversarial Networks (GANs) in the life sciences has been their use as the main instruments for the analysis of biological images [6], [7], [13], [14]. They accomplish this by facilitating cell datasets augmentation and by making the creation of realistic synthetic images more straightforward. Despite this, their employment as a method for predicting the temporal changes in cells has barely been touched upon [12], [10], [8], [19], [20].

III. Methodology

1) Dataset: The dataset uses a multi-well plate design, with distinct wells in a standardized 96-well plate layout denoted by the spatial coordinates B10, C10, and D10. This configuration enables nature to be imitated by growing the same SHSY5Y cell line in three different wells (B10, C10, and D10) under the same culture conditions.

- Biological Replication: Three different wells (B10, C10, D10) containing the same SHSY5Y cell line under identical culture conditions
- Spatial Control: Accounts for potential positional effects within the plate (e.g., edge effects, temperature gradients)
- Statistical Robustness: Enables cross-validation across different physical locations while maintaining experimental consistency

The temporal sequencing within each well allows the model to learn both spatial and temporal consistencies in cell growth patterns.

A. Dataset and Preprocessing

Using the LiveCell dataset from the Sartorius Cell Instance Segmentation Kaggle competition. It's packed with time-lapse microscopy videos of cell cultures, specifically focusing on neuroblastoma cells (like the SHSY5Y line) captured from three different spots in a multi-well plate: B10, C10, and D10. To keep things straightforward for training and testing, B10 and C10 handle the training data, while D10 is set aside just for testing. Each image file name smartly bakes in the details—starting with "Phase" for the imaging mode, followed by the well location (e.g., B10), and then a timestamp in the format [days]d[hours]h[minutes]m. This makes it super easy to track how the cells evolve over time in each spot.

1) Cell Line Characteristics: The data comprises the SHSY5Y human neuroblastoma cells which are methodology of choice in neuroscience and cancer research. These cells are the products of human bone marrow neuroblastoma and are of the neural crest lineage, thus they are perfect to investigate the mechanisms of regeneration, differentiation, and proliferation in cells.Their behavior, which is well-understood, gives us a firm and realistic ground for making predictions of regeneration patterns.

The dataset contains a total of 528 images. B10, C10, and D10 in this dataset each have 176 images.

2) Preprocessing Pipeline: Our preprocessing pipeline employs computer vision techniques to enhance cellular features and generate training masks:

1) Image Inversion and Enhancement
   - Apply Gaussian blur with 5×5 kernel for noise reduction
   - Invert intensity: $I_{\text{inv}} = 255 - I_{\text{blurred}}$ to highlight cells as bright regions
   - Apply histogram equalization for contrast enhancement
2) Cell Segmentation and Mask Generation
   - Apply Otsu's thresholding for automatic binary segmentation
   - Extract contours using connected component analysis
   - Generate binary masks preserving cellular regions
   - Store masks separately for training and testing sets
3) Temporal Sequence Reconstruction

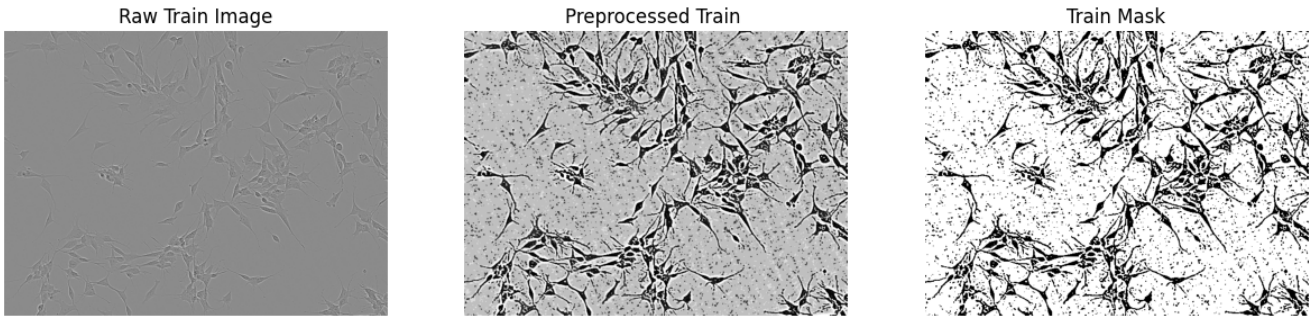


Fig. 1: Using custom python preprocessing pipeline to enhance the raw SH-SY5Y phase-contrast microscopy images which improved the visibility of cellular structures .Comparison between original microscopy image (left), enhanced preprocessed image (middle), and binary mask after segmentation (right) for SHSY5Y_Phase_B10_1_01d08h00m_4_mask.png . The preprocessing pipeline highlights cellular structures while the mask isolates regions of biological interest for focused .

- Parse timestamps from filenames to minute-level precision
- Group images by spatial location and temporal continuity
- Construct input-output pairs: $(X_t, X_{t+1}, X_{t+2}, X_{t+3}) \rightarrow Y_{t+4}$

3) Mathematical Formulation: The preprocessing transformation can be expressed as:

$$I_{processed} = \text{Equalize}(\text{Invert}(\text{GaussianBlur}(I_{original}))) \quad (1)$$

where the inversion operation emphasizes cellular structures:

$$I_{inv}(x, y) = 255 - I_{blurred}(x, y) \quad (2)$$

This is used to invert the colors of the cells.Here, the cells showed white color,so color inversion was applied for the cells in black color.

Temporal sequences are structured to capture cell growth dynamics, with the model learning to predict frame Y_{t+4} from the preceding four frames $\{X_t, X_{t+1}, X_{t+2}, X_{t+3}\}$.

$$I_{norm} = \frac{I - \mu}{\sigma} \quad (3)$$

A sample preprocessed microscopy sequence used for model training is shown in Fig. 1. I stands for the original image, is the mean pixel intensity, and refers to the standard deviation of intensity values. The dataset is in the form of input-output pairs, thus the model gets 4 consecutive frames to predict the next one .

B. Implementation

All models were implemented in Python using TensorFlow 2.x.x. Detailed data exploration and baseline results are available in the public Kaggle notebook. The notebook shows the preprocessing pipeline, including normalization and dataset partitioning for training and validation. It also documents the implementation of baseline architectures used for comparative analysis.

Forecasting Neuroblastoma Cell Growth .

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i) \quad (4)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f) \quad (5)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (6)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o) \quad (7)$$

$$H_t = o_t \circ \tanh(C_t) \quad (8)$$

Fig. 2: The ConvLSTM architecture showing the updation of its cell and hidden states as follows (Equations 4–8).

C. ConvLSTM2D GAN Framework

1) convLSTM: Our Generative Adversarial Network consists of two components. The ConvLSTM architecture used in the proposed framework is illustrated in Fig. 2.

2) Generator: The generator G takes a sequence of 4 frames and produces a future frame: Generator is shown in Fig. 3

$$G(X_{1:4}) = \hat{Y} \quad (4)$$

3) Discriminator: The discriminator D distinguishes between real cell frames and generated ones:

$$D(Y) = \begin{cases} 1 & \text{if } Y \text{ is real} \\ 0 & \text{if } Y \text{ is generated} \end{cases} \quad (5)$$

The training objective follows the minimax game:

$$\min_G \max_D \mathbb{E}[\log D(Y)] + \mathbb{E}[\log(1 - D(G(X)))] \quad (6)$$

Discriminator is shown in Fig. 3

IV. Results

A. Training Strategy

We employ a three-phase training approach:

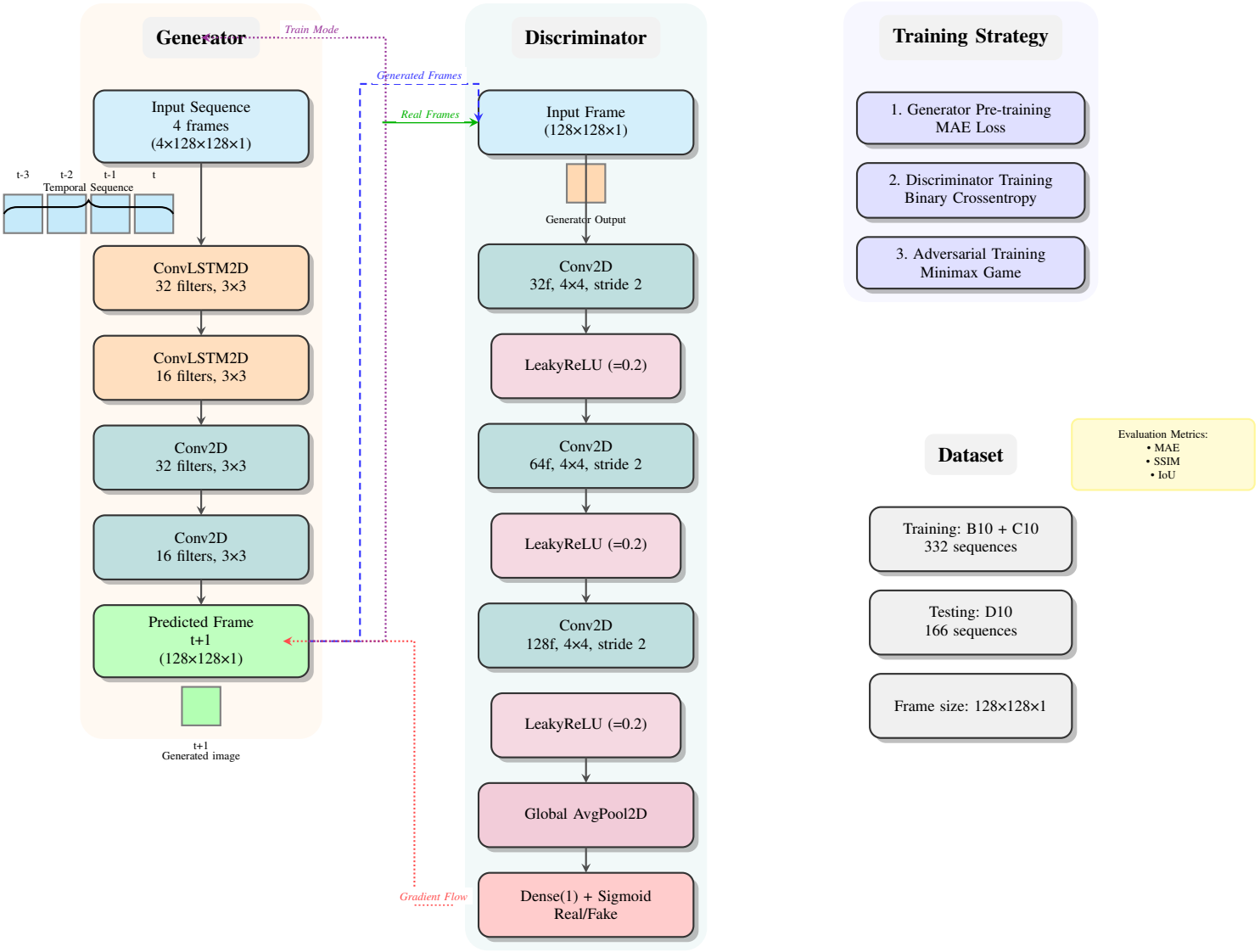


Fig. 3: **Enhanced ConvLSTM-GAN Architecture for Cell Growth Prediction.** The generator processes temporal sequences through ConvLSTM and convolutional layers to predict frame t+1. The discriminator classifies frames as real or generated. The architecture includes a comprehensive testing phase comparing generated frames against ground truth using multiple metrics (MAE, SSIM, IoU) and discriminator evaluation.

1) Generator Pre-training: Train generator alone using mean absolute error loss.
2) Discriminator Training: Train discriminator to distinguish real from generated frames.
3) Adversarial Training: It refers to the combined training of the generator and the discriminator.

B. Experimental Setup

All experiments were conducted using TensorFlow 2.x with Adam optimizer (learning rate = 2×10^{-4}). The models were trained for 200 epochs with batch size of 8.

C. Dataset Split Strategy

A sequence-based split was employed to evaluate generalization across different experimental conditions:

- Training: Sequences B10_1, B10_2, C10_1, C10_2 (332 samples)
- Testing: Sequences D10_1, D10_2 (166 samples)
- Validation: 20% of training data held out

This split strategy tests the model's ability to generalize to unseen experimental conditions (D10) rather than just unseen frames from the same sequences.

D. Evaluation Metrics

Evaluating model performance using:

- Precision: Accuracy of positive predictions.
- Recall: Completeness of event detection.
- Structural Similarity Index (SSIM): Measures perceptual similarity between predicted and actual regeneration pat-

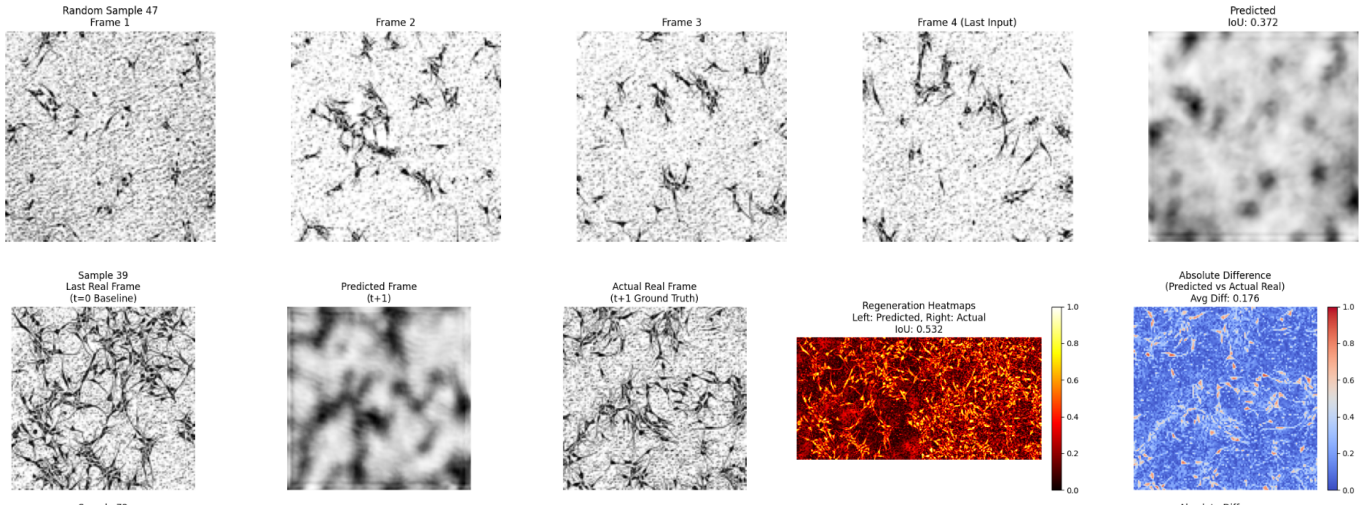


Fig. 4: Model prediction after using first 4 frames to predict the next frame with comparison and comparison graph(green for true positive, red for false positive, blue for false negative).

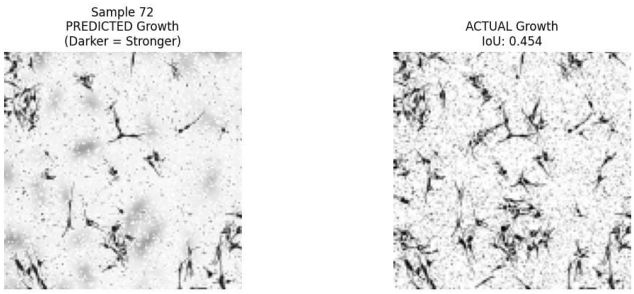


Fig. 5: Comparison between predicted growth and actual growth of neuroblastoma cells.

terns. Unlike pixel-wise metrics (MAE, MSE), SSIM evaluates whether the spatial structure and shape of regenerated cell regions match the ground truth, making it more aligned with biological relevance

- IoU: is a measure of the extent to which the predicted image overlaps with the actual one, thus providing a definite score of segmentation accuracy

E. Obtained Results

The result of final predicted image in Fig. 5 is obtained by setting a threshold of 0.15 using grayscale imaging from the blurred generated image.Lighter shade in the predicted image is the learning rate of the frames 2,3 and 4 from the input. The model learns frame 1 the most due to which the darkened regions appear to be similar to frame 1.The light grayed regions are due to learning of frame 2,3 and 4.Model accuracy is shown in table II. The level of learning for each frame is given in fig. 6 and table I.

F. Interpretability Analysis

Our occlusion analysis revealed temporal importance patterns shown in table I.

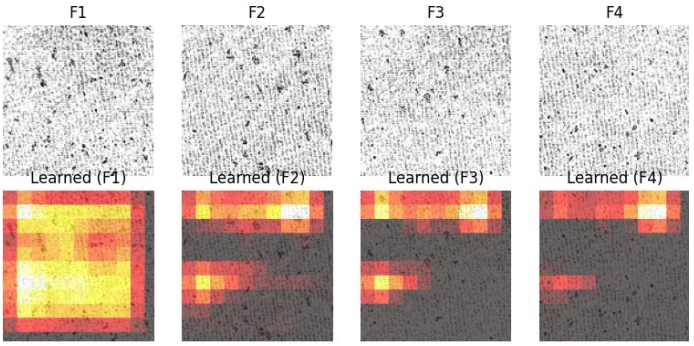


Fig. 6: Occlusion analysis Map

The images in Fig. 6 shows the area of region the model is trying to learn,the more colour it radiates to yellow ,the more it learns.It shows how much each frame is being learnt

TABLE I: Occlusion Impact Analysis on Frame Prediction

Frame	Occlusion Impact (MAE)
Frame 1	0.3578
Frame 2	0.0595
Frame 3	0.0511
Frame 4	0.0301

In simpler terms, Frame 1 was nearly 12 times more influential than Frame 4 in predicting cell growth outcomes.

Spatially, the right-side regions showed 1.65× greater importance than the left side, pointing to clear asymmetric growth patterns in the dataset in table

The image in fig.6 shows the area of region the model is trying to learn,the more colour it radiates to yellow ,the more it learns.It shows how much each frame is being learnt

G. Quantitative Results

Biological Interpretation: The ConvLSTM–GAN model maintains morphological balance (0.90× symmetry ratio) from fig. 4 and concentrates learning on biologically active regions. Approximately half of the total cell-growth zone is consistently emphasized, suggesting the model's strong alignment with genuine cellular regrowth dynamics. IoU , precision and recall is mentioned in the below table II.

TABLE II: Random Sample Performance Summary

Metric	Mean	Std
IoU	0.487	0.071
Precision	0.763	0.049
Recall	0.635	0.092

SSIM is 0.0601 because exact pixel-to-pixel correspondence is difficult to achieve in temporal microscopy prediction.

H. Temporal Importance Patterns

The disproportionate importance of early frames suggests that initial cell states contain sufficient information to predict future growth patterns. This, in turn, has significant implications for the experimental design as it may give researchers the opportunity to make trustworthy predictions from shorter observation periods.

I. Spatial Asymmetry

The stronger influence of features on the right side—while possibly a quirk of training—might actually reflect real biological asymmetries in how cells grow. More work is needed to figure out if this is a true biological pattern or just a bias in the model.

J. Conclusion and Future Work

Using the LiveCell dataset which is part of the Sartorius Cell Instance Segmentation Kaggle competition, ConvLSTM and GAN models can be used as tools for predicting cell growth. Our approach yields strong prediction outcomes and also reveals distinct, interpretable patterns of the main factors that influenced the space and time. It is really impressive how the very first frames weigh so much in accurate predictions thus pointing to ways of simplifying cell biology experiments.

In fact, this study is essentially opening a very convenient path for artificial intelligence application in the prediction of cellular behavior with a promise to eventually speed up drug discovery, toxicity testing, cancer research, and regenerative medicine.

Although the present model is affected by imaging conditions and variability of the cell type, the next research will be dedicated to transfer learning for model adaptation to different cell lines, adding biologically informed constraints, broadening to multi-step forecasting instead of single-frame prediction, coupling with mechanistic biophysical models, and creating realistic synthetic image generation for robustness and applicability enhancement.

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